

SHORT COMMUNICATION

Pediatric
OBESITY

WILEY

Contemporary screen time usage among children 9–10-years-old is associated with higher body mass index percentile at 1-year follow-up: A prospective cohort study

Jason M. Nagata¹  | Puja Iyer¹ | Jonathan Chu¹ | Fiona C. Baker^{2,3} | Kelley Pettee Gabriel⁴ | Andrea K. Garber¹ | Stuart B. Murray⁵ | Kirsten Bibbins-Domingo⁶ | Kyle T. Ganson⁷

¹Division of Adolescent and Young Adult Medicine, Department of Pediatrics, University of California, San Francisco, San Francisco, California, USA

²Center for Health Sciences, SRI International, Menlo Park, California, USA

³Department of Physiology, School of Physiology, University of the Witwatersrand, Johannesburg, South Africa

⁴Department of Epidemiology, University of Alabama at Birmingham, Birmingham, Alabama, USA

⁵Department of Psychiatry and Behavioral Sciences, University of Southern California, Los Angeles, California, USA

⁶Department of Epidemiology and Biostatistics, University of California, San Francisco, San Francisco, California, USA

⁷Factor-Inwentash Faculty of Social Work, University of Toronto, Toronto, Ontario, Canada

Correspondence

Jason M. Nagata, Division of Adolescent and Young Adult Medicine, Department of Pediatrics, University of California, San Francisco, 550 16th Street, 4th Floor, Box 0110, San Francisco, CA 94158, USA. Email: jason.nagata@ucsf.edu

Funding information

American Heart Association Career Development Award, Grant/Award Number: CDA34760281; National Institutes of Health, Grant/Award Numbers: K23 MH115184, K24DK103992

Summary

Objective: There is a paucity of prospective research exploring the relationship among contemporary screen time modalities (e.g., video streaming, video chatting, texting and social networking) and body mass index (BMI) percentile. The objective of this study was to determine the prospective associations between screen time behaviours in a large and demographically diverse population-based cohort of 9–10-year-old children and BMI percentile at 1-year follow-up.

Methods: We analyzed prospective cohort data from the Adolescent Brain Cognitive Development (ABCD) Study ($N = 11\,066$). Multiple linear regression analyses were conducted to estimate associations between baseline screen time behaviours (exposure) and BMI percentile at 1-year follow-up, adjusting for race/ethnicity, sex, household income, parent education, depression, binge-eating disorder and baseline BMI percentile.

Results: Each additional hour of total screen time per day was prospectively associated with a 0.22 higher BMI percentile at 1-year follow-up (95% CI 0.10–0.34) after adjusting for covariates. When examining specific screen time behaviours, each additional hour of texting ($B = 0.92$, 95% CI 0.29–1.55), video chat ($B = 0.72$, 95% CI 0.09–1.36) and video games ($B = 0.42$, 95% CI 0.06–0.78) was significantly prospectively associated with higher BMI percentile.

Conclusions: Screen time is prospectively associated with a higher BMI percentile 1 year later among children 9–10 years old.

KEYWORDS

adolescents, body mass index, obesity, paediatrics, screen time, smart phone, social media, television, weight

1 | INTRODUCTION

The number of children and adolescents presenting with obesity has significantly increased in recent years, with more than 13 million youth ages 2–19 in the United States having been affected by

childhood obesity.¹ Understanding the modifiable factors associated with childhood obesity is key in providing tailored education prevention, and interventions for youth and their families. Excessive screen time use (over 2 h per day) is associated with greater sedentary time, less physical activity and unhealthy eating behaviours, all of which

may contribute to youth's risk for obesity.²⁻⁴ The American Academy of Paediatrics has recommended limitations of technology usage in youth less than 6 years of age.² Studies initially targeted TV-watching and obesity,^{4,5} but only recently included video games.⁶ However, children across the nation are interacting with technology in rapidly evolving ways, and technology is quickly emerging as the premier mode for youth to engage with their peers, family members, education and entertainment.

Cross-sectional studies have demonstrated an association between screen time and higher body mass index (BMI) and obesity,^{4,6-8} with larger studies focused on older adolescents and young adults.⁹ The few prospective studies have been limited to relatively small sample sizes^{10,11} and have not included large, diverse and representative child populations in the United States. Given that the biggest rise in obesity prevalence is seen in late childhood to early adolescence, research investigating BMI and screen time in younger adolescents is vital in identifying emerging trends in paediatric obesity.¹² In addition, most of the existing literature pertains to undifferentiated total screen time or takes a narrower focus on television as the primary medium of media consumption.^{4,5} Although youth interact with a diverse array of screens, there is sparse research exploring BMI's relationship to newer modalities such as texting, social networking, video chatting and video streaming. The purpose of our study, therefore, is to determine prospective associations with several contemporary forms of screen time and BMI percentile in a large, diverse sample of American youth. We hypothesized that contemporary screen time modalities would be prospectively associated with a higher BMI percentile.

2 | METHODS

2.1 | Study population

The Adolescent Brain Cognitive Development (ABCD) Study, a large, diverse and prospective cohort study of brain development and health across adolescence, provided data for this analysis. The sample consisted of 11 875 children recruited from 21 sites around the United States. We analyzed data from the ABCD 3.0 release for the baseline (2016–2018, ages 9–10 years) and 1-year follow-up (2017–2019, ages 10–11 years) assessments. Further details of the study sample, recruitment process, procedures and measures have been previously reported.¹³ Participants with missing data for baseline screen time ($N = 63$) or BMI at 1-year follow-up ($N = 752$) were excluded, leaving a total of 11 066 in the cohort. We used single imputation by Gaussian normal regression for missing covariate data ($N = 1016$) based on other available sociodemographic data as we have done previously.¹⁴ Centralized institutional review board (IRB) approval was obtained from the University of California, San Diego. Study sites obtained approval from their local IRBs. Caregivers provided written informed consent and each child provided written assent.

2.2 | MEASURES

2.2.1 | Exposures: Screen time

Screen time was measured using the ABCD Youth Screen Time Survey. Children answered questions about typical hours per day spent on six different screen modalities (viewing/streaming television shows or movies, watching/streaming videos [e.g., YouTube], playing video games, texting, video chatting [e.g., Skype, Facetime] and social networking [e.g., Facebook, Instagram, Twitter])¹⁵ separately for weekdays and weekend days based on a previously validated measure.¹⁶ Similar to a previous study investigating screen time exposure from the ABCD study, we calculated the weighted average participants' typical weekday and weekend screen time usage to obtain a typical week measure.¹⁷ After obtaining this screen time total for each type of media, we then reported the weighted average as a continuous variable.

2.2.2 | Outcome: BMI percentile

BMI was calculated based on the average of two-to-three measured heights and weights ($BMI = \text{weight}/\text{height}^2$) by research staff and was converted into sex and age-specific percentiles in accordance with Center for Disease Control and Prevention growth curves and definitions.¹⁸

2.2.3 | Covariates

Age, sex, race/ethnicity, household income and highest parent education were based on parents' self-report. Major depressive disorder and binge-eating disorder were assessed using the Kiddie Schedule for Affective Disorders and Schizophrenia (KSADS-5) parent diagnostic interview.¹⁹

2.3 | Statistical analysis

Data analysis was performed in 2020 using Stata 15.1. Multiple linear regression analyses were conducted to estimate the association between baseline screen time (exposure variables, examined in separate models) and BMI percentile at 1-year follow-up (outcome variable), adjusting for baseline BMI percentile only (model 1), as well as baseline BMI percentile, sex, race/ethnicity, household income, parent education and site (model 2). Some children within the sample were twins or siblings. Sensitivity analyses were conducted, including only one sibling per family and findings did not differ; therefore, we present results from the full sample. We also analyzed BMI and BMI z-score as outcomes. We tested for effect modification of screen time by sex, but there was no evidence of effect modification (p for interaction >0.05). ABCD's pre-constructed propensity weights were applied based on the American Community Survey from the US Census.²⁰

TABLE 1 Sociodemographic, screen time and anthropometric characteristics of 11 066 Adolescent Brain Cognitive Development (ABCD) study participants

	Mean (SD)/%
<i>Sociodemographic characteristics (baseline)</i>	
Age (years)	9.9 (0.6)
Sex (%)	
Female	48.8%
Male	51.2%
Race/ethnicity	
White	52.2%
Latino/Hispanic	20.0%
Black	17.3%
Asian	5.5%
Native American	3.2%
Other	1.9%
Household income	
Less than \$25 000	18.7%
\$25 000 through \$49 999	20.4%
\$50 000 through \$74 999	17.5%
\$75 000 through \$99 999	13.4%
\$100 000 through \$199 999	22.6%
\$200 000 and greater	7.4%
Parent with college education or more	79.7%
Major depressive disorder	0.2%
Binge-eating disorder	0.7%
<i>Screen time variables (hours per day, baseline)</i>	
Total screen time	3.99 (3.16)
Television shows/movies	1.31 (1.31)
Videos (e.g., YouTube)	1.05 (1.18)
Video games	1.06 (1.13)
Texting	0.24 (0.56)
Video chat	0.21 (0.52)
Social networking	0.13 (0.45)
<i>Anthropometrics</i>	
BMI percentile, baseline	62.1 (30.7)
BMI z-score, baseline	0.43 (1.49)
BMI (kg/m ²), baseline	19.02 (4.30)
BMI category, baseline	
Underweight (BMI < 5th percentile)	3.9%
Normal weight (5th percentile ≤ BMI < 85th percentile)	62.5%
Overweight (85th percentile ≤ BMI < 95th percentile)	15.9%
Obese (BMI > 95th percentile)	17.8%
BMI percentile, 1-year follow-up	63.5 (30.5)
BMI z-score, 1-year follow-up	0.48 (1.54)
BMI (kg/m ²), 1-year follow-up	20.02 (4.77)
BMI category, 1-year follow-up	

(Continues)

TABLE 1 (Continued)

	Mean (SD)/%
Underweight (BMI < 5th percentile)	3.8%
Normal weight (5th percentile ≤ BMI < 85th percentile)	60.8%
Overweight (85th percentile ≤ BMI < 95th percentile)	16.6%
Obese (BMI > 95th percentile)	18.9%

Note: ABCD propensity weights were applied based on the American Community Survey from the US Census.

Abbreviation: SD, standard deviation.

3 | RESULTS

Table 1 describes sociodemographic characteristics of the 11 066 participants included. The sample was balanced by sex (48.8% female) and racially and ethnically diverse (47.8% non-White). On average, youth reported 4 h of screen time per day, with the most time spent watching/streaming television shows/movies (1.3 h), playing video games (1.1 h) and watching/streaming videos (1.1 h) at baseline. Mean BMI percentile was 62.1 at baseline and 63.5 at 1-year follow-up.

Table 2 shows linear regression analyses that examine the prospective associations between baseline screen time and BMI percentile at 1-year follow-up. In models adjusting for baseline BMI percentile only, all forms of screen time were associated with higher BMI percentile at 1-year follow-up. In fully adjusted models, each additional hour of total screen time per day was prospectively associated with a 0.22 higher BMI percentile at 1-year follow-up (95% CI 0.10–0.34). When examining specific screen time behaviours, each additional hour of texting ($B = 0.92$, 95% CI 0.29–1.55), video chat ($B = 0.72$, 95% CI 0.09–1.36) and video games ($B = 0.42$, 95% CI 0.06–0.78) was significantly prospectively associated with a higher BMI, after adjusting for covariates.

Appendices A and B show sensitivity analyses with BMI and BMI z-score as outcomes, respectively. The findings are similar (all B coefficients with positive association), although the scales are different and some associations with individual screen time modalities are no longer statistically significant.

4 | DISCUSSION

In a population-based, demographically diverse cohort of 9–11-year-old children in the United States, we found that greater screen time was prospectively associated with a higher BMI percentile 1 year later. When examining specific screen time behaviours, more time spent on texting, video chat and video games was most strongly associated with a higher BMI percentile. Although texting and watching/streaming videos had the strongest associations with BMI, it should be noted that participants only reported an average of less than 15 min per day on these screen time modalities.

Our findings extend upon those of mostly cross-sectional studies in older adolescents or adults examining the relationship between screen time

TABLE 2 Prospective associations between baseline screen time and body mass index (BMI) percentile at 1-year follow-up in the Adolescent Brain Cognitive Development study

	BMI percentile at follow-up, adjusted for baseline BMI percentile only		BMI percentile at follow-up, fully adjusted ^a	
	<i>B</i> (95% CI)	<i>p</i>	<i>B</i> (95% CI)	<i>p</i>
Total screen time	0.29 (0.18–0.40)	<0.001	0.22 (0.10–0.34)	<0.001
Television	0.46 (0.15–0.76)	0.003	0.32 (0.02–0.62)	0.039
Videos (YouTube)	0.58 (0.29–0.87)	<0.001	0.36 (0.06–0.67)	0.019
Video games	0.53 (0.20–0.86)	0.001	0.42 (0.06–0.78)	0.021
Texting	1.25 (0.62–1.87)	<0.001	0.92 (0.29–1.55)	0.004
Video chat	0.98 (0.34–1.61)	0.003	0.72 (0.09–1.36)	0.025
Social networking	1.04 (0.29–1.80)	0.007	0.67 (–0.07–1.41)	0.075

Note: Bold indicates $p < 0.05$. The *B* coefficient in the cells represents abbreviated output from a series of linear regression models with BMI percentile as the dependent variable and screen time as the independent variable. Thus, the table represents the outputs from 14 regression models in total. ABCD propensity weights were applied based on the American Community Survey from the US Census.

^aCovariates: race/ethnicity, sex, household income, parent education, depression, binge-eating disorder, site and baseline BMI percentile.

and BMI.^{4,6–8} We add to the prior literature on screen time and BMI by reporting prospective associations from a large, demographically diverse cohort study of children in the United States and study contemporary forms of screen time, including texting, video chatting and social networking.

Several mechanisms may contribute to the prospective association between screen time and higher BMI, including snacking, sedentary time and physical inactivity. Youth may be more prone to snacking and overeating while distracted on screens, even in the absence of hunger.⁸ Children may also be exposed to more food advertisements while on their screens,²¹ thus influencing their eating behaviours. Screen time is mostly sedentary and may also displace time for physical activity.²²

Despite the important findings of this study, limitations should be noted. Although we adjusted for several potential confounders, including baseline BMI percentile, there is the possibility of additional unmeasured confounders. Despite the use of a longitudinal cohort study, given the observational study design, we cannot definitively establish causality. Future research could examine the reverse relationship, which is whether baseline BMI percentile is prospectively associated with screen time at 1-year follow-up. Although BMI percentiles are the standard of care in paediatric clinical practice,^{18,23} there are limitations in the use of BMI percentile in longitudinal studies and at >97th percentile of BMI^{24,25}; thus, we report BMI and BMI z-scores in supplemental appendices. Furthermore, effect sizes of the associations between screen time and BMI percentile were relatively small for 1-year follow-up; however, cumulative exposure to screen time over decades may become more clinically significant later in the life course. In addition, screen time increases after ages 9–11 years, indicating that future research is needed to follow the ABCD cohort through later adolescence. While other reports have similarly found relatively low usage of social networking among pre-adolescents 8–12-years-old (16 min per day), adolescents 13–18-years-old report a much higher usage of social networking (71 min per day).²⁶ Lastly, screen time measures were based on self-report, which could lead to reporting bias.

Findings from this study suggest that specific screen time behaviours should be targeted as part of childhood obesity prevention efforts. Today's youth are increasingly engaging in screen time for

social and educational purposes, which has been exacerbated by the COVID-19 pandemic.²⁷ This has led to a rapid rise in overall screen time use that can have detrimental effects. Guidance for children, parents, clinicians and policymakers regarding appropriate screen time usage is needed to ensure the health and well-being of young people.²⁷ Health care providers should assess for excess screen time usage and weight changes, especially during the pandemic, and advise about potential risks associated with excessive sedentary screen time. Professional organizations, such as the American Academy of Pediatrics, should provide further specific guidance for families regarding screen time usage and strategies to prevent weight gain related to excessive screen time usage.²

ACKNOWLEDGEMENTS

The authors thank Samuel Benabou and Kelsey Wu for editorial assistance. J.M.N. was supported by the American Heart Association Career Development Award (CDA34760281). S.B.M. was supported by the National Institutes of Health (K23 MH115184). K.B.D. is supported by the National Institutes of Health (K24DK103992). The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript. The ABCD study was supported by the National Institutes of Health and additional federal partners under award numbers U01DA041022, U01DA041025, U01DA041028, U01DA041048, U01DA041089, U01DA041093, U01DA041106, U01DA041117, U01DA041120, U01DA041134, U01DA041148, U01DA041156, U01DA041174, U24DA041123 and U24DA041147. A full list of supporters is available at <https://abcdstudy.org/federal-partners/>. A listing of participating sites and a complete listing of the study investigators can be found at <https://abcdstudy.org/principal-investigators.html>. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in analysis or writing of this report.

CONFLICT OF INTERESTS

The authors have no conflict to declare.

AUTHOR CONTRIBUTIONS

JMN, PI, JC, KPG, KBD, KTG conceived the study. JMN and JC carried out the analysis. PI conducted the literature review. JMN, PI, JC and KTG were involved with writing the article. All authors were involved in critically revising the article and had final approval of the submitted and published versions.

ORCID

Jason M. Nagata  <https://orcid.org/0000-0002-6541-0604>

REFERENCES

- Sanyaolu A, Okorie C, Qi X, Locke J, Rehman S. Childhood and adolescent obesity in the United States: a public health concern. *Glob Pediatr Health*. 2019;6:2333794X19891305. <https://doi.org/10.1177/2333794X19891305>
- Chassiakos YR, Radesky J, Christakis D, et al. Children and adolescents and digital media. *Pediatrics*. 2016;138(5):e20162593. <https://doi.org/10.1542/peds.2016-2593>
- Lissak G. Adverse physiological and psychological effects of screen time on children and adolescents: literature review and case study. *Environ Res*. 2018;164:149-157. <https://doi.org/10.1016/j.envres.2018.01.015>
- Stiglic N, Viner RM. Effects of screentime on the health and well-being of children and adolescents: a systematic review of reviews. *BMJ Open*. 2019;9(1):e023191. <https://doi.org/10.1136/bmjopen-2018-023191>
- Robinson TN. Reducing children's television viewing to prevent obesity: a randomized controlled trial. *Jama*. 1999;282(16):1561-1567. <https://doi.org/10.1001/jama.282.16.1561>
- Falbe J, Willett WC, Rosner B, Field AE. Body mass index, new modes of TV viewing and active video games. *Pediatr Obes*. 2017;12(5):406-413. <https://doi.org/10.1111/ijpo.12158>
- Cameron JD, Maras D, Sigal RJ, et al. The mediating role of energy intake on the relationship between screen time behaviour and body mass index in adolescents with obesity: the HEARTY study. *Appetite*. 2016;107:437-444. <https://doi.org/10.1016/j.appet.2016.08.101>
- Hicks K, Pitts SJ, Lazorick S, Fang X, Rafferty A. Examining the association between screen time, beverage and snack consumption, and weight status among Eastern North Carolina Youth. *N C Med J*. 2019;80(2):69-75. <https://doi.org/10.18043/ncm.80.2.69>
- Kenney EL, Gortmaker SL. United States adolescents' television, computer, videogame, smartphone, and tablet use: associations with sugary drinks, sleep, physical activity, and obesity. *J Pediatr*. 2017;182:144-149. <https://doi.org/10.1016/j.jpeds.2016.11.015>
- Neufeld ND. Outcome analysis of the B.E. S.T.R.O.N.G. childhood obesity treatment program: effectiveness of an eight-week family-based childhood obesity program using an internet-based health tracker. *Child Obes*. 2016;12(4):227-236. <https://doi.org/10.1089/chi.2015.0048>
- Oellinrath IM, Svendsen MV. BMI-specific associations between health-related behaviours and overweight - a longitudinal study among Norwegian adolescents. *Public Health Nutr*. 2017;20(3):481-491. <https://doi.org/10.1017/S1368980016002536>
- Hales CM, Carroll MD, Fryar CD, Ogden CL. Prevalence of obesity among adults and youth: United States, 2015-2016; 2017.
- Barch DM, Albaugh MD, Avenevoli S, et al. Demographic, physical and mental health assessments in the adolescent brain and cognitive development study: rationale and description. *Dev Cogn Neurosci*. 2018;32:55-66. <https://doi.org/10.1016/j.dcn.2017.10.010>
- Nagata JM, Iyer P, Chu J, et al. Contemporary screen time modalities among children 9-10years old and binge-eating disorder at one-year follow-up: a prospective cohort study. *Int J Eat Disord*. 2021;54(5):887-892. <https://doi.org/10.1002/eat.23489>
- Bagot KS, Matthews SA, Mason M, et al. Current, future and potential use of mobile and wearable technologies and social media data in the ABCD study to increase understanding of contributors to child health. *Dev Cogn Neurosci*. 2018;32:121-129. <https://doi.org/10.1016/j.dcn.2018.03.008>
- Sharif I, Wills TA, Sargent JD. Effect of visual media use on school performance: a prospective study. *J Adolesc Health*. 2010;46(1):52-61. <https://doi.org/10.1016/j.jadohealth.2009.05.012>
- Guerrero MD, Barnes JD, Chaput JP, Tremblay MS. Screen time and problem behaviors in children: exploring the mediating role of sleep duration. *Int J Behav Nutr Phys Act*. 2019;16(1):105. <https://doi.org/10.1186/s12966-019-0862-x>
- Barlow SE, the Expert Committee. Expert Committee recommendations regarding the prevention, assessment, and treatment of child and adolescent overweight and obesity: summary report. *Pediatrics*. 2007;120(Supplement 4):S164-S192. <https://doi.org/10.1542/peds.2007-2329C>
- Townsend L, Kobak K, Kearney C, et al. Development of three web-based computerized versions of the kiddie schedule for affective disorders and schizophrenia child psychiatric diagnostic interview: preliminary validity data. *J Am Acad Child Adolesc Psychiatry*. 2020;59(2):309-325. <https://doi.org/10.1016/j.jaac.2019.05.009>
- Heeringa S, Berglund P. A guide for population-based analysis of the adolescent brain cognitive development (ABCD) study baseline data. *bioRxiv*. 2020. <https://doi.org/10.1101/2020.02.10.942011>
- Harris JL, Kalnova SS. Food and beverage TV advertising to young children: measuring exposure and potential impact. *Appetite*. 2018;123:49-55. <https://doi.org/10.1016/j.appet.2017.11.110>
- Gunnell KE, Flament MF, Buchholz A, et al. Examining the bidirectional relationship between physical activity, screen time, and symptoms of anxiety and depression over time during adolescence. *Prev Med*. 2016;88:147-152. <https://doi.org/10.1016/j.ypmed.2016.04.002>
- Grossman DC, Bibbins-Domingo K, Curry SJ, et al. Screening for obesity in children and adolescents: US Preventive Services Task Force recommendation statement. *Jama*. 2017;317(23):2417-2426. <https://doi.org/10.1001/jama.2017.6803>
- Cole TJ, Faith MS, Pietrobelli A, Heo M. What is the best measure of adiposity change in growing children: BMI, BMI %, BMI z-score or BMI centile? *Eur J Clin Nutr*. 2005;59(3):419-425. <https://doi.org/10.1038/sj.ejcn.1602090>
- Berkey CS, Colditz GA. Adiposity in adolescents: change in actual BMI works better than change in BMI z score for longitudinal studies. *Ann Epidemiol*. 2007;17(1):44-50. <https://doi.org/10.1016/j.annepidem.2006.07.014>
- Rideout V, Robb M. *The Common Sense Census: Media Use by Tweens and Teens*. San Francisco, CA: Common Sense Media; 2019:1-104. <https://www.common-sense-media.org/research/the-common-sense-census-media-use-by-tweens-and-teens-2019>
- Nagata JM, Abdel Magid HS, Gabriel KP. Screen time for children and adolescents during the coronavirus disease 2019 pandemic. *Obesity*. 2020;28(9):1582-1583. <https://doi.org/10.1002/oby.22917>

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

How to cite this article: Nagata JM, Iyer P, Chu J, et al.

Contemporary screen time usage among children 9-10-years-old is associated with higher body mass index percentile at 1-year follow-up: A prospective cohort study. *Pediatric Obesity*. 2021;16(12):e12827. <https://doi.org/10.1111/ijpo.12827>